

CHAINLAB

Distributed Ledgers and Decentralization

Networks, trust assumptions and governance

FREE BLOCKCHAIN EDUCATION RESOURCE

This guide is designed for learning and revision. It explains foundational concepts and highlights practical limitations and safety considerations.

Learning objectives

- Distinguish distributed databases from blockchains.
- Identify different dimensions of decentralization.
- Evaluate trust assumptions in a network.

1. Distributed ledger basics

- A distributed ledger is a record replicated across multiple participants. A blockchain is one way to organize and update such a ledger; not every distributed ledger uses blocks or a chain.
- Replication can improve availability and shared visibility, but it introduces coordination, consistency and operational challenges.

2. Dimensions of decentralization

- Infrastructure: how many independent machines and operators participate.
- Consensus: who can propose or confirm state changes and how power is distributed.
- Development: whether one team controls protocol changes or multiple groups contribute.
- Governance: how upgrades and disputes are decided.
- Access: whether participation is permissionless or controlled by an administrator.

3. Trust models

- A permissioned system may trust identified operators and legal agreements. A permissionless system relies more heavily on public protocol rules, economic incentives and open verification.
- Users still rely on software clients, interfaces, bridges, custodians, RPC providers and governance processes. Decentralization is therefore a spectrum, not a binary guarantee.

4. Governance and upgrades

- Protocol changes may be coordinated through developers, validators, token holders, foundations, companies or community processes.
- Governance mechanisms can create legitimacy and coordination, but can also concentrate influence or leave important stakeholders underrepresented.

5. Evaluation checklist

- Ask who can censor transactions, change rules, halt service, upgrade software or control critical infrastructure.
- Check validator/operator concentration, client diversity, geographic distribution, entry requirements and the practical ability to exit or fork.

Review questions

- Why is decentralization multidimensional?
- How can an application depend on centralized infrastructure even if its base chain is decentralized?
- What governance questions should a user investigate?

Educational notice: This material is for general education only. It is not financial, legal, tax or investment advice. Verify current network and project details before acting.